absorbs energy in the visible region of light and thus imparts blue colour to the solution. The solutions are paramagnetic and on standing slowly liberate hydrogen resulting in the formation of an amide.

$$M^+ + e^- + NH_3 \rightarrow MNH_2 + \tfrac{1}{2}H_2$$

In concentrated solution, the blue colour changes to bronze colour and become diamagnetic.

Reaction with water:

Alkali metals react with water to give corresponding hydroxides with the liberation of hydrogen.

$$2\ Li + 2\ H_2O \rightarrow 2\ LiOH + H_2$$

They also react with alcohol, and alkynes which contain active hydrogens.

$$2\ Na + 2\ C_2H_5OH \rightarrow 2\ C_2H_5ONa + H_2$$

$$H\text{-}C \equiv C\text{-}H \xrightarrow{Na} H\text{-}C \equiv C\text{-}Na \xrightarrow{Na} Na\text{-}C \equiv C\text{-}Na$$

Reducing activity:

Alkali metals can lose their valence electron readily hence they act as good reducing agents.

$$M_{(s)} \rightarrow M^+_{(g)} + e^-$$

Reaction with carbon:

Lithium directly reacts with carbon to form the ionic compound, lithium carbide. Other metals do not react with carbon directly. However, when they are treated with compounds like acetylene they form acetelydes.

$$2\ Li + 2C \rightarrow Li_2C_2$$

5.2.4 Uses of alkali metals:

i. Lithium metal is used to make useful alloys. For example with lead it is used to make 'white metal' bearings for motor engines, with aluminium to make aircraft parts, and with magnesium to make armour plates. It is used in thermonuclear reactions.

ii. Lithium is also used to make electrochemical cells.

iii. Sodium is used to make Na/Pb alloy needed to make $Pb(Et)_4$ and $Pb(Me)_4$. These organolead compounds were earlier used as anti-knock additives to petrol, but nowadays lead-free petrol in use.

iv. Liquid sodium metal is used as a coolant in fast breeder nuclear reactors.

v. Potassium has a vital role in biological systems.Potassium chloride is used as a fertilizer.

vi. Potassium hydroxide is used in the manufacture of soft soap. It is also used as an excellent absorbent of carbon dioxide.

vii. Caesium is used in devising photoelectric cells.

5.3 General characteristics of the compounds of alkali metals

All the common compounds of the alkali metals are generally ionic in nature. General characteristics of some of their compounds are discussed here.

11th Chemistry Chapter-5 Vol-1 EM.indd 131 09-01-2023 10:54:34